SPE-229302-MS

Digital Twin for Pipeline Leak Monitoring

M. Hamilton, Computer Science, Memorial University, St. John's, NL, Canada; H. Ferroudji, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar; M. Colbourne and S. Sheppard, Birch Scientific, St. John's, NL, Canada; A. Barooah, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar; A. K. Sleiti, Mechanical Engineering, Qatar University, Doha, Qatar; I. Hassan, Petroleum Engineering Department, Texas A&M University at Qatar; M. S. Khan, Interdisciplinary Research Center for Membranes and Water Security, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia; S. Rezaei-Gomari, Teeside University, Middlesbrough, UK; M. A. Rahman, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

We demonstrate how a visual digital twin system is used to implement a digital twin for pipeline monitoring. A visual digital twin allows for ingestion of data for calibration and creation of models which enable a digital replica of a real-world physical system for decision support and improving situational awareness. A laboratory experimental pipeline system is used to generate data to drive modelling capabilities in this system. We show how machine learning operations (MLOps) principles are applied in the context of digital twins, forming a sub-study area known as Digital Twin ML Ops (DT MLOps). The intended purpose of the system is to both train operators of the leak detection system in its use and provide high situational awareness and operational readiness to users.

We demonstrate how multiple sources of monitoring data from an experimental pipeline setup, as well as simulation can be combined in a visual digital twin system and used for pipeline leak detection and leak plume and flow visual prediction. We show how leak detection and visual leak prediction are visualized in the context of a pipeline twin along with confidence and uncertainty and other explainable elements from machine learning models. We demonstrate how models are tracked using the DT MLOps sub system.

The overall system demonstrates a novel combination of real experimental data driving models for both leak detection and the prediction of the associated leak plume and pipeline flow visual assessment imagery. This demonstrates a system which can detect leaks and their locations and also provide operators with assessment of the leak. The system provides provenance through its DT MLOps capabilities. This presented virtual pipeline and leak model, which integrates AI with experimental validation, is not widely available in industry.

Introduction

Leak detection in pipelines represents an on-going challenge in various industries, notably offshore oil and gas. Leaks in offshore pipelines represent a particular risk given the sensitive and challenging environment, presenting immediate environmental and safety issues. Detecting leaks as early as possible and responding to prevent further environmental damage, contain existing leakage and neutralize associate safety issues is a critical requirement for sustainable and legal operations.

Fluid flow, in particular multiphase flow, can exhibit complex behaviors under a variety of flow conditions (Barooah, Khan, et al. 2024; Ferroudji, Saad Khan, et al. 2025). Leaks modify normal flow behavior but can be difficult to detect as their influence on already complex behavior itself can be nuanced (Ferroudji et al. 2024). Pipeline geometry required to accommodate non-trivial seafloor routes add further challenge. In a damaged pipeline, it is possible that multiple leaks may further conflate and complexify detection and localization of leaks. The development of leak detection models has been an ongoing subfield and source of challenge (W. Al-Ammari, Sleiti, Hamilton, et al. 2025; Barooah, Saad Khan, et al. 2024; Ferroudji, Khan, Barooah, Rahman, et al. 2025; Harati et al. 2024; Jujuly et al. 2016; M S Khan et al. 2024; Muhammad Saad Khan et al. 2025; Rahman et al. 2024).

Managing pipeline integrity thus becomes a complex activity encompassing monitoring and response. Several sensing modalities can be employed to aid in leak monitoring and response, including acoustic monitoring, visual inspection, pressure monitoring, chemical tracers, etc. In terms of leak response, further modelling and understanding the surrounding environment is also relevant. All of this sits in the context of a pre-construction design, which is embodied in a dynamic as-built state, where streams of monitoring data from various modalities are being collected and available. All of the associated models and data results in what is known as a digital thread, which is a term describing the flow of all information associated with a system being modelled (Zhang, Liu, and Chen 2024). In the context of pipeline monitoring, the digital thread could encompass a wide range of information from modelling and simulation of a design pre-construction, operational specifications to temporal streams of pipeline and environmental mapping and monitoring data. This digital thread is complex and composed of disparate data elements composed of different data forms from different modalities. These data elements are often isolated, in that they are stored separately and represented differently using varying file formats. This can be a practical consequence of designing a data collection system using sensors from different vendors and/or modalities where ultimately different standards can pervade.

For a digital thread, what is needed is an analytics system that enables integration of knowledge from all these sources into a common framework that enables a human user or perhaps an artificial intelligence agent to use this potentially dense digital thread for decision support and situational awareness. The concept of a digital twin provides an approach to meeting these requirements. A digital twin is a virtual, dynamic representation of a physical system. A digital twin system seeks to integrate digital thread into a cohesive unit. However, to provide value to a human operator, a digital twin system must provide an effective interface. A visual twin then is a visual representation of a digital twin (M. Hamilton et al. 2023) and provides a necessary interface to enable understanding of the complex integrative digital twin system.

Digital twins typically combine live data from the digital thread with system models to calibrate the underlying model to better reflect the twinned physical system. This fits naturally with data-driven modelling approaches. In recent years, significant advances in machine learning techniques have resulted in impressive physical models which are largely data-driven. Data-driven models are suited for complex phenomenon which do not readily admit closed-form or human-specified modelling forms. While black-box machine learning models often can be validated as providing good accuracy in representing certain phenomenon, the lack of explainability can be a downside. In the context of decision support for human operators, understanding the why behind machine-learning driven analytics can be of significant value when high stakes decisions must be made in relation to potential leak or other potentially serious pipeline events.

Pipeline operators must monitor a potentially massive digital thread comprised of complex sensing data and design specifications. In the event of a pipeline leak, they must act decisively to prevent environmental, safety and economic disaster. Interpreting a wide swath of monitoring data continuously represents a task of significant cognitive load for a human. Combining artificial intelligence and presenting data in a more digestible form to human operators can potentially enable improved situational awareness of the pipeline status while reducing massive cognitive load of interpreting changing conditions. This can improve operator confidence as well as promote sound and quicker decision making in the event of catastrophic leaks or anomalies.

What is desired is a digital twin system which provides an ability to integrate and enhance the digital thread with additional insights provided via conventional analytics, signal processing and data mining as well as modern machine learning and AI techniques. In the context of pipeline monitoring, we desire to integrate pipeline design and operating flow specifications. The digital twin system must also provide visual twin interface which enables rapid and effective digestion of information by human operators from the raw data which is collected. In this paper, we explore how to build such a system and its use in pipeline monitoring.

The contributions of the paper are as follows. We present a digital twin system for pipeline monitoring including a visual twin based on gaming engine technology. The digital twin system has integrated machine learning operations capabilities which enables centralized tracking, version control, comparison and reproducibility for trained machine learning models. We call this Digital Twin Machine Learning Ops (DT MLOps). We demonstrate pipeline monitoring using a laboratory experimental pipeline. We demonstrate integration of a digital thread comprised of data from acoustic sensors, pressure sensors, flow visualization camera, and high-speed leak plume visualization. We demonstrate how this thread can be represented in the visual twin and augmented by insights from machine learning models derived from the digital thread. In particular, we demonstrate how the visual twin integrated with DT MLOps can prove to be an effective analysis tool during machine learning development, as model predictions in the visual twin can be represented in the visual twin.

Related Work

Leak detection and pipeline analysis methods have been explored extensively, for conventional oil and gas transport scenarios as well as in transport of water. Many techniques for leak detection have been proposed, from analysis techniques based on wavelets to data-driven techniques based on machine learning (Ferroudji, Khan, Barooah, Al-Ammari, et al. 2025; Harati et al. 2024, 2024; Jujuly et al. 2016; Kim et al. 2021; Rahmes et al. 2015; Sinha et al. 2020; Yang, Zhao, and Rourke 2019). Leak behavior can also be viewed under the large range of techniques designed for anomaly detection (Bouman, Bukhsh, and Heskes 2023). A recent review gives a comprehensive overview of leak detection techniques (Rahman et al. 2024).

Pipeline monitoring, however, is a broader topic that requires large context. Leak detection is critical, however understanding the larger context of a pipeline system, its design and disaster response procedures become relevant in addition to the key task of detecting leaks. Digital twins provide a framework whereby a pipeline system can be replicated digitally and thus analyzed and monitored digitally as well (Tao et al. 2019). Pipeline related digital twins have been explored previously (W. A. Al-Ammari, Sleiti, Hamilton, et al. 2025), with comprehensive review available (W. A. Al-Ammari, Sleiti, Rahman, et al. 2025). A visual twin provides an additional level of usability to a strict digital twin, allowing for better user experience and operator effectiveness, supporting decision-making (M. Hamilton al. 2023). A pipeline leak visual twin was presented previously, however it only considered simulated pipeline data (M. Hamilton et al. 2023). Similar visual capabilities were explored previous in a virtual arctic simulation environment (Matthew Hamilton and Dodd 2015) with integrated data visualization with possibility for physical simulation in the context of subsea arctic energy operations. A follow-on work introduced integration of flow assurance analysis into the

VASE (Matthew Hamilton 2016), showing how pipeline geometry design could be aided in an integrated environment. Bhowmik also presented a digital twin (Bhowmik 2019) for pipeline leak detection.

Generally, the integration of visual twin, streaming sensing data and production machine learning development and deployment techniques form the field of MLOps has not been extensively presented nor available readily in industry.

Methods, Procedures and Process

Digital Twin System Overview

Digital twin systems enable incorporation of the digital thread relevant to the object, system or process being twinned. This includes real-time, live sensing data streams. The visual twin is driven mainly by a gaming engine which enables 3D visualization and simulation (M. Hamilton et al. 2023). In our system, static or fixed elements of the digital thread may be incorporated as assets into the gaming engine, whereas numeric, text and otherwise dynamic sensing data are ingested via a standard application protocol interface (API) mechanism and stored in a database (See Fig 1).

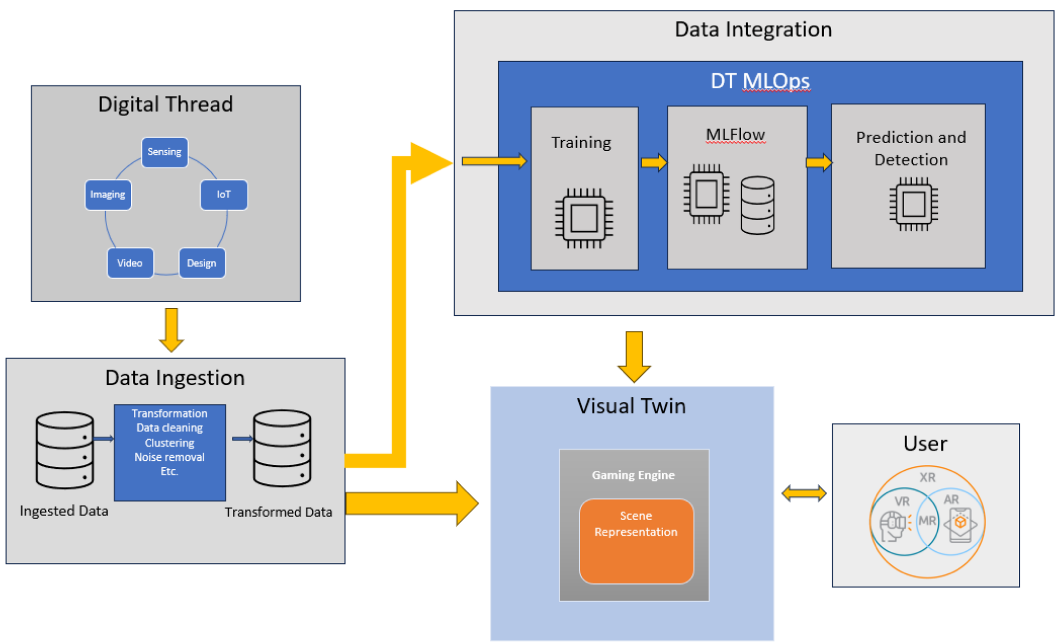

Figure 1—Diagram of digital twin system design.

Data Analytics and Digital Twin Machine Learning Ops (DT MLOps)

Historic or static data streams can be stored in a database. This data is ideal for training machine learning models that can be applied to live streaming data in the future for analytics and predictive modeling which will support operator situational awareness and decision-making. This can be particularly useful if historic data has been labelled and augmented with human-expert knowledge.

Machine learning generally involves an initial step of exploratory data analysis which informs preprocessing. We can develop appropriate custom preprocessing that can be applied to both historical and on-going streaming data, creating new processed data streams (See Fig 1). Custom preprocessing modules can be integrated into the system and re-used as needed for ongoing processing of raw data streams to processed ones (transformed data as illustrated in Fig. 1).

Machine learning generally requires multiple iterations of the process of 1) implement, 2) experiment then 3) comparative analysis (Cawley and Talbot 2010). Software such as MLflow (mlflow.org) can enable a convenient tracking of these iterations and supports analysis interfaces which enable the subsequent comparative analysis phase. In the presented digital twin system, we train machine learning models from processed streams, which are tracked in MLflow as part of a per-experiment logging. In addition to the data, the exact code commits which contribute to an experiment are stored along with other parameters required to make all experiments reproducible.

Machine learning models can then be trained directly from these processed streams. We support "no-code" operation here by enabling users to choose conventional machine learning models (e.g. those supported by the popular ML library Scikit-learn) or specify custom models.

Machine learning models typically perform better when trained on the most extensive dataset which covers the target domain best. Ongoing data collection suggests that new scenarios may emerge which are represented in new data that could be integrated into a newer training set for a machine learning model. Careful experimental logging is required to manage this requirement for continuous model evolution considering newly available training data. The intent of incorporating new data is model improvement, however this must be quantified and validated through a suite of metrics. It is required that new models improve on performance of older models and do not degrade performance. A variety of metrics must be considered, stored and presented to a model engineer using this system as improvement vs. degradation cannot necessarily be decided appropriately due to a single metric. For example, model biases may creep in due to bad training data or training data from a significantly different distribution.

We consider the Data Integration module depicted in the overall system diagram Fig. 1 and its components in Fig. 2. All of the experimental tracking information is available immediately to the visual twin and can be brought in for visualization in the context of the twin's scene (more on this in the case studies below). This can enable model engineers to perform contextual comparative analysis of models, in addition to enabling predictions to be used as helpful analytic overlaps in the visual twin.

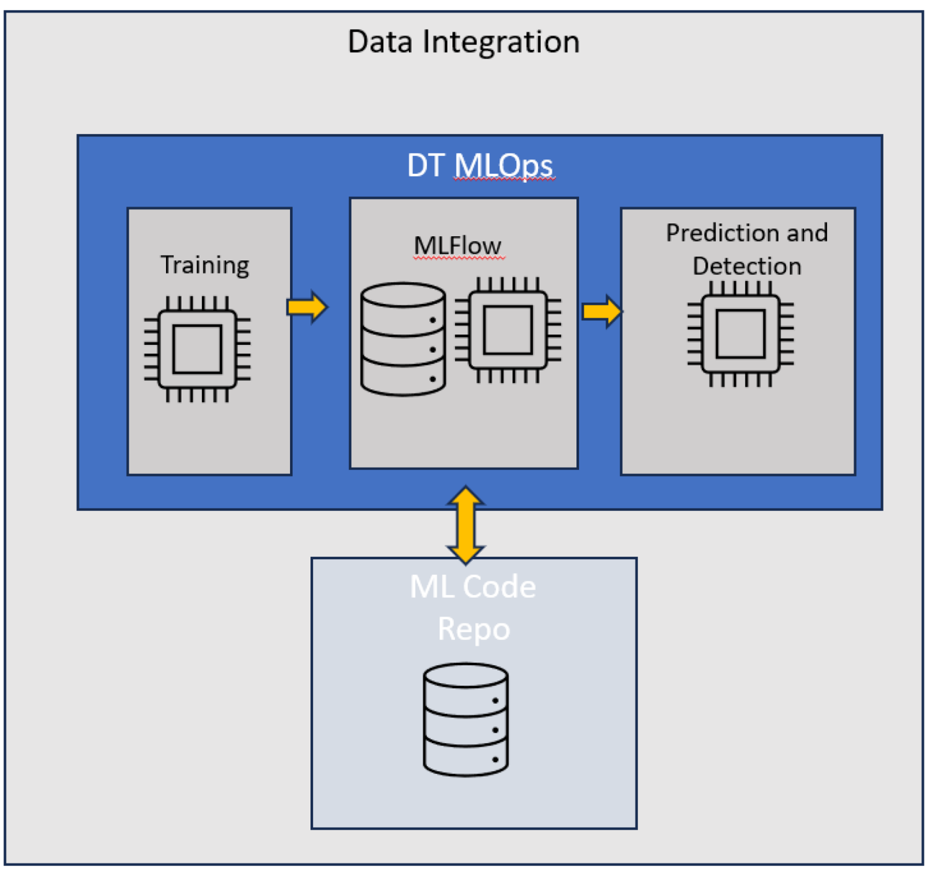

Figure 2—Data Integration module diagram.

Pipeline Monitoring Digital Twin

We demonstrate the use of the described digital twin system in the context of a pipeline digital twin designed to support pipeline monitoring. For these purposes, we employ a laboratory experimental pipeline flow loop. We first describe this experimental setup. We use data collected from a laboratory experimental pipeline system (pipeline length: 6 m, leak sizes: 3 mm, 2.5 mm and 1.8 mm, spaced 90 mm). The leaks can be enabled and disabled. We consider the cases of 1, 2 and 3 leaks. The data models leak in a variety of flow settings, at a maximum operating pressure for the mixture of two 2 bar and with air and water velocity mixture velocity in the pipe of 1.5 m/s. This system was described previously in the context of leak detection method development (Ferroudji et al. 2025) and depicted below graphically in Fig. 3.

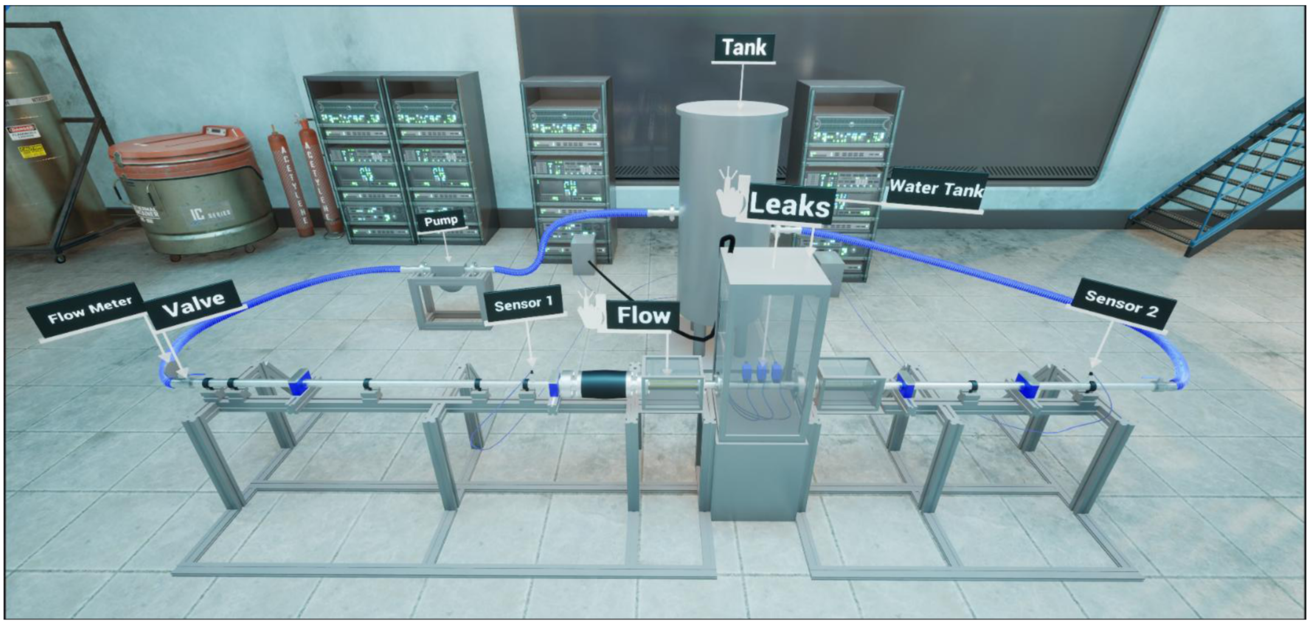

Figure 3—Diagram of experimental leak system.

We collected data from pressure sensors, acoustic sensors, high speed cameras capturing pipeline flow and leak imagery and incorporated it as training data into a digital twin system to train leak detection models and pipeline flow and leak imagery prediction models. Data was collected starting from a common initial start time, but individual elements of the digital thread have different sampling rates and time spans. In this investigation, water and air were considered as the liquid and gas phases, respectively. The tank was filled with water to a volume of 200 L before the beginning of the experiment. Subsequently, the liquid phase was introduced into the pipeline from the tank by using a centrifugal slurry pump, and the gas phase was injected via a mass flow rate controller.

The steps considered in the experiments were as follows:

1. Preparation of recording devices, including dynamic sensors, flowrate sensors, high-speed camera (for bubble visualization in the water tank), and DSLR camera (for flow regime map visualization in the main pipeline).
2. Calibration of the measuring devices.
3. Liquid phase introduction in the main pipeline
4. Gas-phase injection via mass flow rate controller.
5. Leaks opened for 30s after the development of the flow regime in the main pipeline (approximately 7s after the start of the experiment).
6. Recording of various parameters.

7. Repeat over a range of different superficial liquid and gas velocities, with 1, 2, and 3 leaks.

To guarantee that the liquid phase flowed through the flow loop system at different flow rates (from 134 kg/min to 323.5 kg/min), a slurry pump was placed after the tank. A mass controller is placed at the pipeline intake introducing the gas phase into the system. Additionally, several instruments were used to measure the parameters to determine the dynamic pressure, flow rate, and pressure gradient: a dynamic pressure sensor, Coriolis flow meter, and pressure transducer.

The water tank had dimensions of 0.5m × 0.5m × 1m and it was made of transparent acrylic with a thickness of 20 mm. A high-speed camera was used to record the motion of the bubbles in the aquarium to observe the flow through the leaks and obtain greater insight into the behavior of leaks while taking underwater circumstances into account. In addition, a hydrophone was inserted inside the tank to collect hydroacoustic signals during leaks. Table 1 summarizes the characteristics of the flow-loop system.

Table 1—Characteristics of the flow loop system.

Parameter	Characteristics and properties
Pipeline length	6 m
Pipeline Diameter	50.8 mm
1st leak diameter	3 mm
2nd leak diameter	2.5 mm
3rd leak diameter	1.8 mm
Distance between leaks	90 mm
Distance to the 1st leak	3457 mm
Water tank dimensions	0.5 × 0.5 × 1 m
Pipeline material	Stainless-steel
Water tank material	Transparent acrylic
Liquid phase density	$\rho_w = 998\ kg/m^3$
Liquid phase viscosity	$\mu_w = 0.001\ kg/(m{\cdot}s)$
Gas phase density	$\rho_A = 1.2\ kg/m^3 - 2.4\ kg/m^3$
Gas phase viscosity	$\mu_A = 1.7894e - 05\ kg/(m{\cdot}s)$

We see a schematic of the experimental setup in Fig. 3. We employ several cameras in this setup in order to collect video data. The water tank in Fig 3 contains several controllable leaks. A Photron FASTCAM high speed camera is placed around 50cm from the water tank and aimed at the leak modules (See figure for example image). Video is captured at 1000 Hz over a period of 3 seconds. For flow visualization, a DSLR camera is used to capture the clear flow pipeline at a location prior to the water tank (see figure).

In the digital twin, the design of the laboratory was modelled in 3D for the purpose of providing context for visualization (see bare lab figure), capturing the key elements of the experimental setup. We show some of the information from Table 1 in 3D context. Greater spatial insight is provided when this is viewed interactively with camera movement and is enhanced further when viewed using augmented, virtual, and mixed reality (e.g. Microsoft Hololens and Varjo XR-4) or holographic displays (e.g. LookingGlass, Avalon Holographic NOVAC), where the provision of additional visual cues (e.g. stereoscopic and focal cues) is known to enhance perception and analytic performance (cite some studies we know).

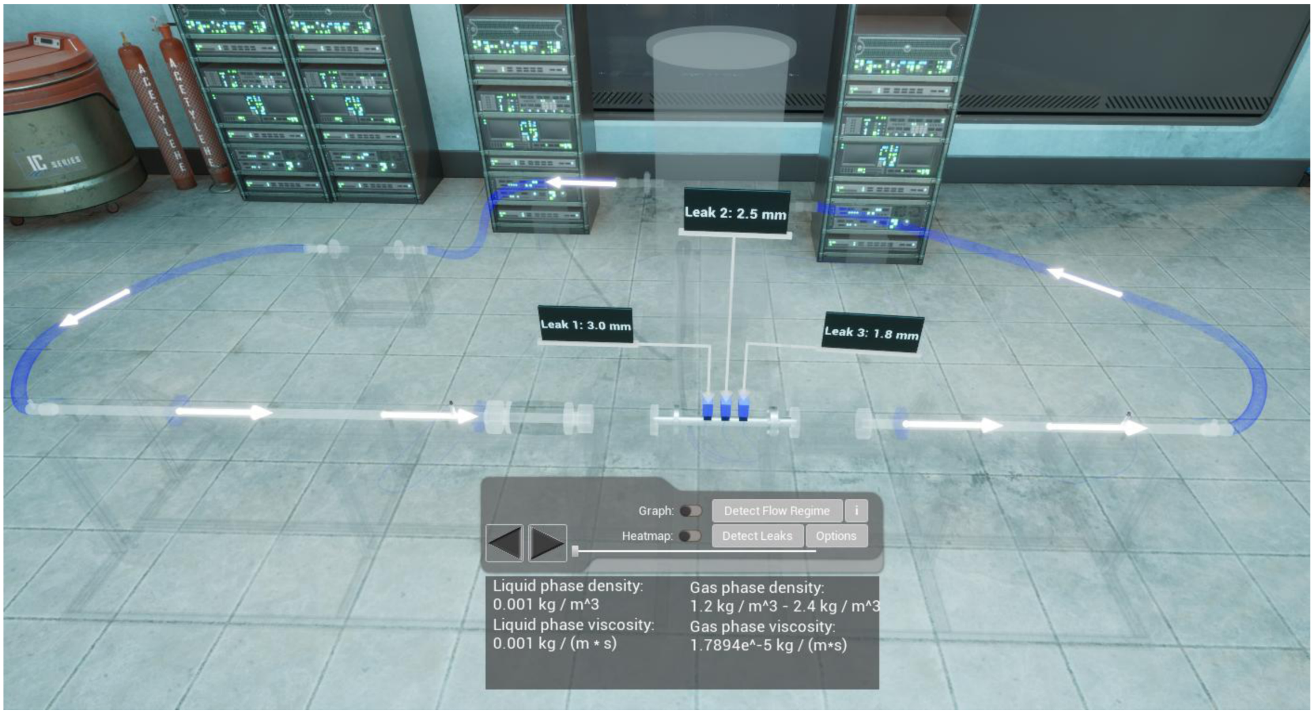

Figure 4—Digital laboratory twin showing flow and leak specifications in 3D environment.

Case Study: Leak detection

We describe the digital twin working with the pipeline digital thread in several case studies. A previous work explored various leak detection methods employed on data collected from the experimental pipeline described above (Ferroudji, Khan, Barooah, Al-Ammari, et al. 2025). Leak detection methods have been explored extensively in other contexts as well, using both experimental and simulation based data (Barooah, Saad Khan, et al. 2024; Ferroudji, Saad Khan, et al. 2025; M. Hamilton et al. 2023; Harati et al. 2024)

In terms of a digital twin, we may train multiple leak detection methods which have effectiveness in different scenarios, or different biases. Multiple methods can be trained and compared in terms of performance metrics (e.g. various forms of expected error estimation).

While leak detection methods have already been developed and improved in the literature, their practical deployment can be complex. Different methods may function better in different flow regimes or it may be most effective to use multiple methods to double monitor a pipeline in order provide a stronger level of leak detection. Furthermore, for data-driven, machine learning approaches, the effectiveness of the method can often be highly-dependent on the training dataset used for the method. Maintaining an effective leak detection system requires juggling all these aspects; different models, different training datasets and potential employment of many combinations thereof simultaneously in order to create the most effective detection analytics tool. The digital twin system with integrated data and DT MLOps enables this complex orchestration.

Case study: Leak Plume Prediction

We use the digital twin to develop a leak plume prediction capability. For training data, we have leak bubble visualization videos captured by the high-speed camera. We first employ object detection computer vision to leak videos to track leak bubbles. This is done using the popular library YOLOv5 (Ultralytics 2025). An example image showing tracking can be seen in Fig W. The resulting tracking data provides a statistical characterization of the bubble movement, known as a bubble vector, which we denote $B(V_{sl}, V_{sg}, l)$,

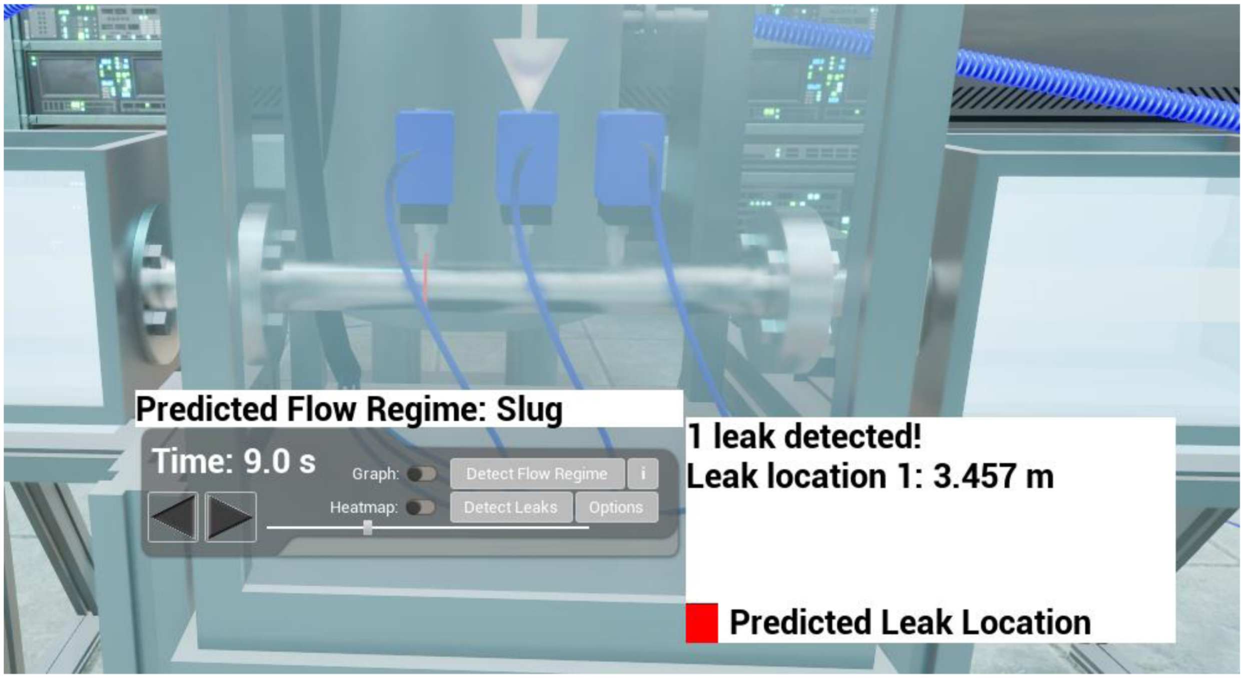

Figure 5—Example leak location prediction visualized with localization error driven from cross-validation experiments from the DT MLOps subsystem.

where V_{sl} is superficial liquid velocity, V_{sg} is superficial gas velocity and l is a vector describing the location of leaks. This characterization is based on bubble size and velocity distributions.

After this preprocessing, the bubble vectors can be stored as processed streams in the database. It is then possible to perform a model development process within the digital twin system. We can formulate a learning task where we learn the function $B(V_{sl}, V_{sg}, l)$ from bubble vector examples. We can use conventional machine learning methods to learn this function based on the implied supervised data or use custom models which will train and run on backend system (Fig. 2).

Results

Integrated Digital Twin

We demonstrate the pipeline monitoring digital twin and its associated visual twin elements. The digital twin allows for interactive use of the 3D rendering capability on 4K conventional display, but also can be readily adapted to immersive displays, both head-worn and glasses-free. Analyzing complex digital twin visualizations with immersive displays shows promise to enable better ability for the user to understand complex relationships in the visual twin, improving cognitive load and decision-making abilities. This topic is worthy of future investigations in the specific context of 3D world based digital twins and for pipeline monitoring

We demonstrate how both the leak bubble and flow map visualization videos can be integrated into the pipeline digital twin. Fig. 6 shows a single frame of the flow map video embedded and aligned with the virtual representation of the equivalent laboratory pipeline. These kinds of automatic registrations and alignments can be readily performed using ChArUco Boards or other calibration patterns along with a calibrated camera, these being conventional operations in computer vision available from the OpenCV library (opencv.org).

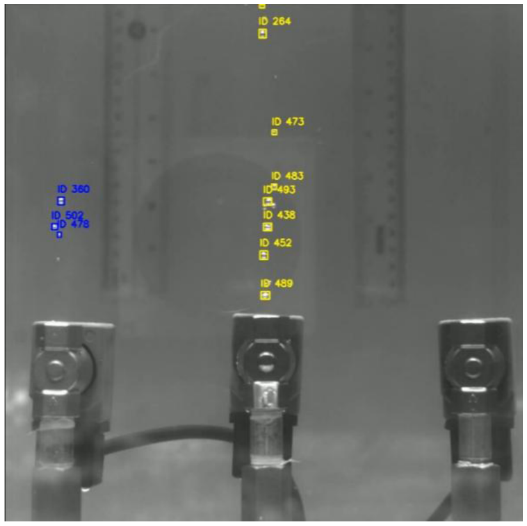

Figure 6—Object tracking using YOLO to determine bubble statistics.

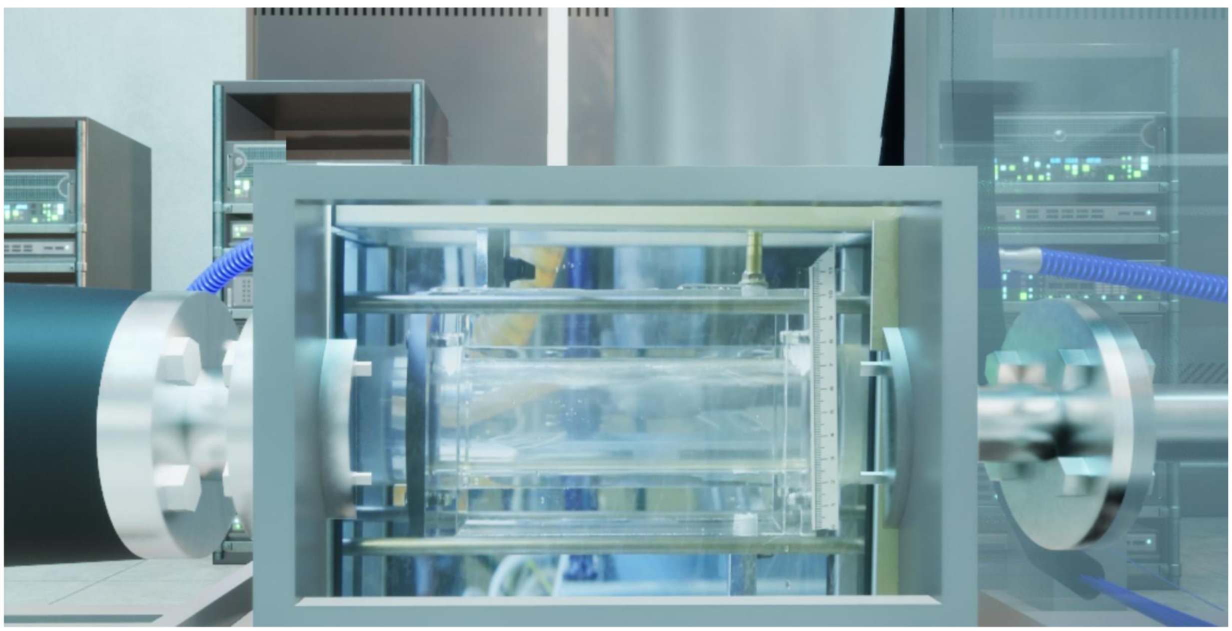

Figure 7—Digital twin pipeline with 2D video overlay. We see virtual and real pipelines registered in the 3D world.

In Fig. 8, we see a single frame of both the bubble and flow visualizations side-by-side. These can be viewed side by side and are shown in the context of their locations in overall virtual laboratory representation. We also demonstrate further integrated visualization capabilities of the twin. In Fig 9, pressure sensor data can be visualized. The data timeline can be adjusted by the time bar present, with rewind and forward capabilities.

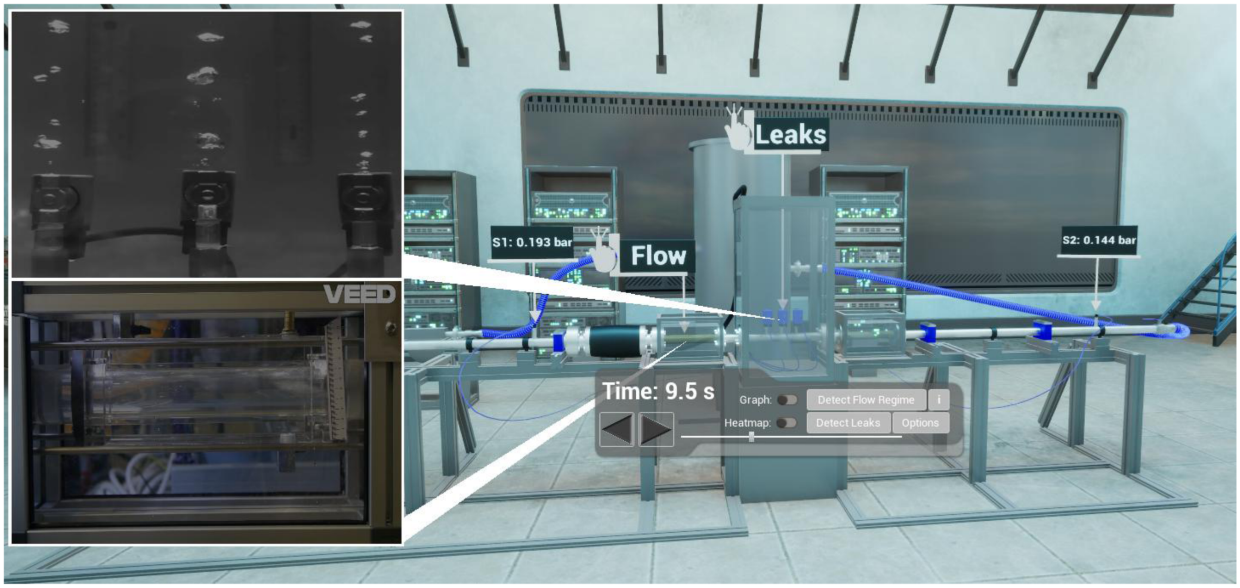

Figure 8—Callouts windows (left) provide zoomed in view, bringing smaller scale scene details into a perceptible range while still showing the "bigger picture" context of the laboratory setup.

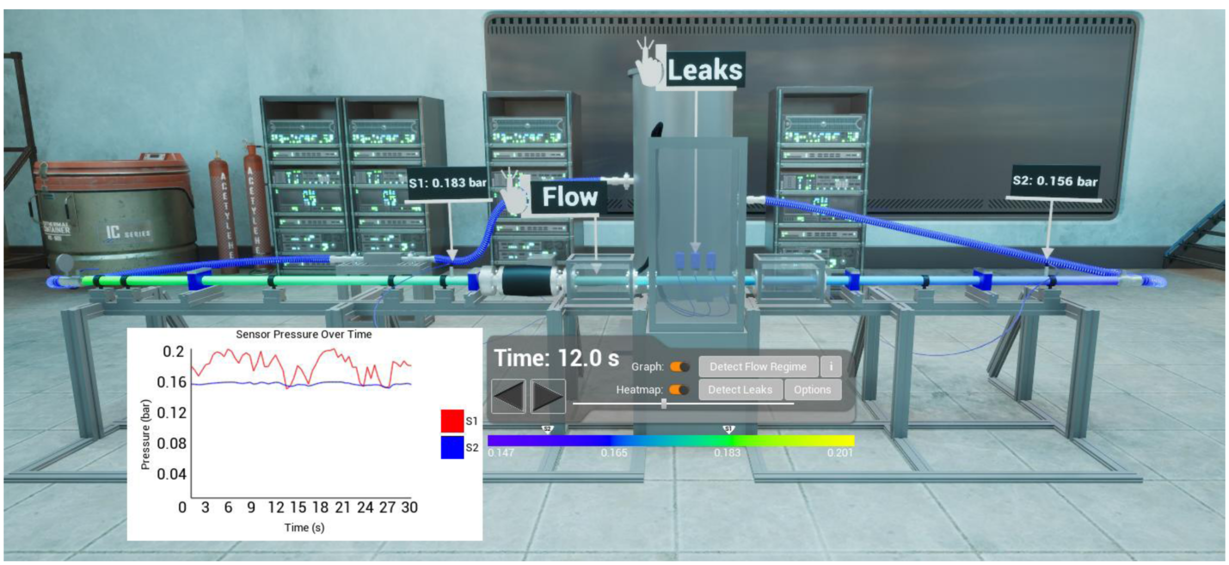

Figure 9—Laboratory twin with labels. Images shows time plot of pressure sensor data along with color map representations.

Integrated Machine Learning Prediction

We demonstrate bubble prediction capabilities based on a machine learning model. We integrated into the gaming engine 3D virtual laboratory scene representation a method to synthesize bubbles in a replica water tank (Fig. 12). This synthesizer takes the bubble vector statistical description of bubble movement and creates bubbles which match its distribution. The synthesizer module simply requires a bubble vector. We can easily train machine learning models to predict a bubble vector from a leak and gas and water velocity and leak location, quantity profile once this information is available in the processed streams database of the digital twin system.

This demonstrates an ability to predict the physical dynamics of leaks, based on a data-driven approach. It is suggested that similar approaches, perhaps requiring further attention to detail, could enable plume dynamic prediction methods for real-world pipeline operations. Combined into the visual twin representation, this would enable operators to have an intuitive view into how leak plumes may propagate in the context of a real-world, as-built 3D world representation.

Figure 10—Bubble synthesis in leak tank based on statistical bubble vector input.

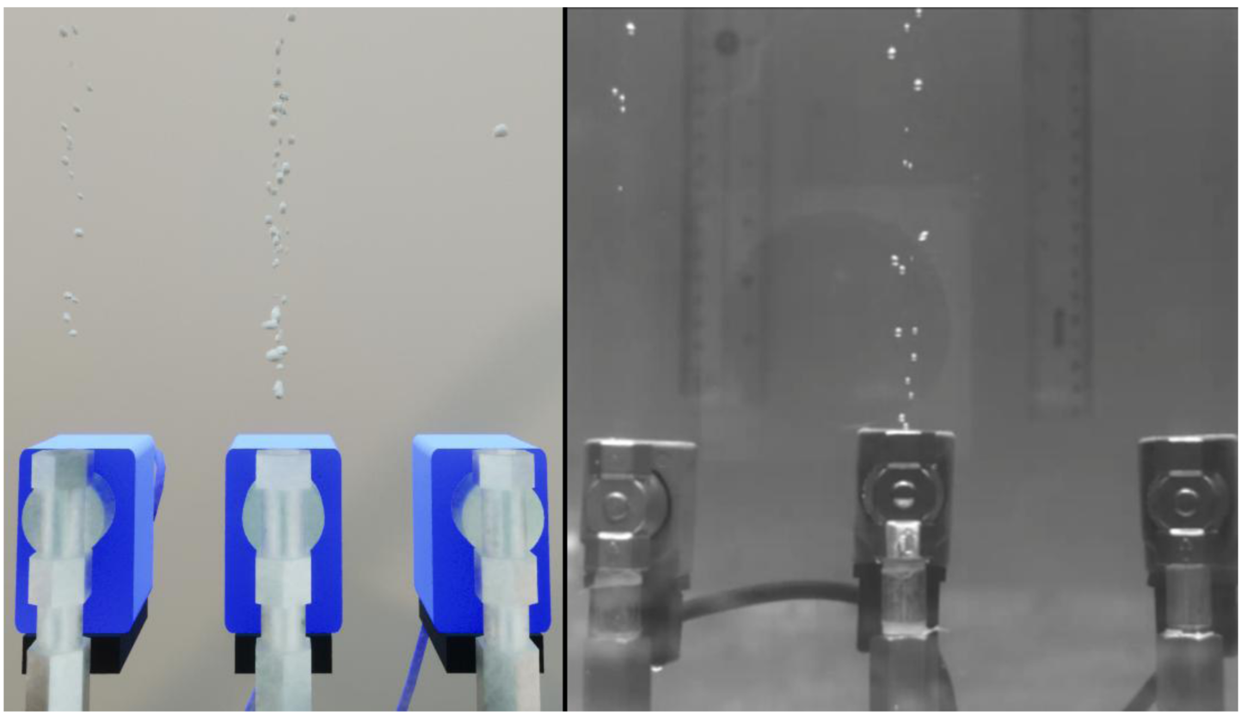

Figure 11—Synthesized bubbles (left) generated from a statistic vector vs. real bubble video (right) based on flow.

Conclusion and Future Work

In the energy industry, pipeline operators face the problem of monitoring pipeline systems, with the risk of failure due to leaks potentially causing significant economic, environmental and safety issues. Sensing devices can be applied to pipeline systems to better model pipeline operations, with the hope that early detection of leaks can enable a better response to mitigate the above risks. Modelling pipeline systems, however, is quite complex. Sensing strategies produce large datasets, but no integrated understanding which helps make decisions when leaks or other operational anomalies occur.

Digital twin systems can solve this issue. They help to integrate the potentially massive digital thread associated with monitoring operations into meaningful analytics and visual representations. This supports more confidence and decisive actions by operators in the event of potentially catastrophic events such as leaks.

In this paper, we have presented a demonstration of a digital twin designed for a pipeline monitoring scenario. We use a laboratory experimental pipeline system as the basis for this digital twin. The digital twin system presented enables pipeline operators to effectively monitor pipelines for leaks by enabling an integrated analysis of the digital thread associated with a pipeline. This integrated data system enables application of leak detection methods and leak plume prediction methods directly from the digital thread data sources. The outputs of these detection and prediction methods can be visualized intuitively using the visual twin, based on high quality gaming engine computer graphics.

In this paper, we have demonstrated these capabilities using an experimental pipeline system. While not the same as a much larger, real-world operating pipeline system, this study suggests that applying the same system on operational pipelines would be feasible. A future prototype would enable operators to both detect leaks immediate, localize them and predict the spread of the resulting leak plume, thus enabling a better understanding of how to respond in such scenarios. Future iterations will integrate further sources of data with improved machine learning models, making predictions based on multi-modal data stream inputs.

Acknowledgement

The work presented in this publication was made possible by NPRP-S grant #NPRP14S-0321-210080 from the Qatar Research, Development and Innovation Council (QRDI). The findings herein reflect the work and are solely the responsibility of the authors.

References

Al-Ammari, W, Ahmad Khalaf\ Sleiti, Matthew Hamilton, et al 2025. "Integration of Unsupervised and Supervised Machine Learning Models for Single/Multiple Leak Detection and Localization Using Minimal Sensor Data." https://aiche.confex.com/aiche/s25/meetingapp.cgi/Home/0.

Al-Ammari, Wahib A., Ahmad K. Sleiti, Mohammad Azizur Rahman, et al 2025. "Digital Twin for Leak Detection and Fault Diagnostics in Gas Pipelines: A Systematic Review, Model Development, and Case Study." *Alexandria Engineering Journal* **123**(March): 91–111. https://doi.org/10.1016/j.aej.2025.03.054.

Al-Ammari, Wahib A, Ahmad K Sleiti, M Hamilton, et al 2025. "AI-Based Adaptive Digital Twin Framework for Real-Time Leak Detection and Localization in Offshore Gas Pipelines." *In International Conference on Offshore Mechanics and Arctic Engineering*,, V006T11A075. https://doi.org/10.1115/OMAE2025-157014.

Barooah, Abinash, Muhammad Saad Khan, et al 2024. "Development of a Mechanistic and Data-Driven Model for Multiphase Flow Leak Detection in Pipeline." *In IPTC International Petroleum Technology Conference*,, D011S028R008. https://doi.org/10.2523/IPTC-24574-MS.

Barooah, Abinash, Muhammad Saad Khan, et al 2024. *INVESTIGATION OF MULTIPHASE FLOW LEAK DETECTION IN PIPELINE USING TIME SERIES ANALYSIS TECHNIQUE*. http://asmedigitalcollection.asme.org/OMAE/proceedings-pdf/OMAE2024/87868/V008T11A047/7362022/v008t11a047-omae2024-127882.pdf.

Bhowmik, Subrata. 2019. "Digital Twin of Subsea Pipelines: Conceptual Design Integrating IoT, Machine Learning and Data Analytics." Proceedings of the Annual Offshore Technology Conference 2019-May.

Bouman, Roel, Zaharah Bukhsh, and Tom Heskes. 2023. "Unsupervised Anomaly Detection Algorithms on Real-World Data: How Many Do We Need?" **25**: 1–34. http://arxiv.org/abs/2305.00735.

Cawley, Gavin C., and Nicola L.C. Talbot. 2010. "On Over-Fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation." *Journal of Machine Learning Research* **11**: 2079–2107.

Ferroudji, Hicham et al 2024. "Numerical Modeling Of Water-Air Multiphase Flow Within A Pipeline In The Presence Of Double Leaks." *International Conference on Fluid Flow, Heat and Mass Transfer* (036): 1–8.

Ferroudji, Hicham, Muhammad Saad Khan, et al 2025. "Analysis of Water-Air Multiphase Flow through a Pipeline with Two Leaks Using Numerical and Experimental Methods." *Journal of Dispersion Science and Technology*.

Ferroudji, Hicham, Muhammad Saad Khan, Abinash Barooah, Mohammad Azizur Rahman, et al 2025. "Analysis of Water-Air Multiphase Flow through a Pipeline with Two Leaks Using Numerical and Experimental Methods." *Journal of Dispersion Science and Technology* **0**(0): 1–12. https://doi.org/10.1080/01932691.2024.2444984.

Ferroudji, Hicham, Muhammad Saad Khan, Abinash Barooah, Wahib A. Al-Ammari, et al 2025. "Comprehensive Analysis of Leak Impacts on Liquid-Gas Multiphase Flow Using Statistical, Wavelet Transform, and Machine Learning Approaches." *Process Safety and Environmental Protection* **194**(December 2024): 825–43. https://doi.org/10.1016/j.psep.2024.12.049.

Hamilton, M. et al 2023. "Visual Twin for Pipeline Leak Detection." In *Society of Petroleum Engineers - ADIPEC, ADIP 2023*, Society of Petroleum Engineers.

Hamilton, Matthew. 2016. *OTC-27392-MS Integrated Design and Analysis for Virtual Arctic Simulation Environment*. http://onepetro.org/OTCARCTIC/proceedings-pdf/16OARC/16OARC/OTC-27392-MS/1352889/otc-27392-ms.pdf/1.

Hamilton, Matthew, and Stephen Dodd. 2015. "Virtual Arctic Simulation Environment." *OTC Arctic Technology Conference 2015* (March): 999–1014.

Harati, Saeed et al 2024. "Performance Analysis of Various Machine Learning Algorithms for CO2 Leak Prediction and Characterization in Geo-Sequestration Injection Wells." *Process Safety and Environmental Protection* **183**: 99–110.

Jujuly, Muhammad, Premkumar Thodi, Aziz Rahman, and Faisal Khan. 2016. "Computational Fluid Dynamics Modeling of Subsea Pipeline Leaks in Arctic Conditions." *Arctic Technology Conference 2016* (December).

Khan, M S et al 2024. "Experimental Investigation of Early Leak Detection in Offshore Multiphase Flow Pipelines Using Non-Invasive Electrical Resistance Tomography." In SPE EOR Conference at Oil and Gas West Asia,, D021S014R001. https://doi.org/10.2118/218536-MS.

Khan, Muhammad Saad et al 2025. "Advancing Offshore Pipeline Safety: Exploring Non-Invasive Electrical Resistance Tomography for Upstream Leak Response Detection Strategy." *Flow Measurement and Instrumentation* **106**: 103013. https://linkinghub.elsevier.com/retrieve/pii/S0955598625002055.

Kim, Juhyun et al 2021. "The Development of Leak Detection Model in Subsea Gas Pipeline Using Machine Learning." *Journal of Natural Gas Science and Engineering* **94**(May): 104134. https://doi.org/10.1016/j.jngse.2021.104134.

Rahman, Mohammad Azizur et al 2024. "Single and Multiphase Flow Leak Detection in Onshore/Offshore Pipelines and Subsurface Sequestration Sites: An Overview." *Journal of Loss Prevention in the Process Industries* **90**(September 2023): 105327. https://doi.org/10.1016/j.jlp.2024.105327.

Rahmes, Mark, Tasha Bingman, Mark Blue, and George Lemieux. 2015. "Continuous Environmentally Efficient Pipeline Leak Detection." *Society of Petroleum Engineers - SPE Canada Heavy Oil Technical Conference 2015, CHOC 2015*: 74–87.

Sinha, Saurabh et al 2020. "Leak Detection in Carbon Sequestration Projects Using Machine Learning Methods: Cranfield Site, Mississippi, USA." Proceedings - SPE Annual Technical Conference and Exhibition 2020-Octob.

Tao, Fei, He Zhang, Ang Liu, and A. Y.C. Nee. 2019. "Digital Twin in Industry: State-of-the-Art." *IEEE Transactions on Industrial Informatics* **15**(4): 2405–15.

Ultralytics. 2025. "YOLOv5." https://github.com/ultralytics/yolov5 (August 7, 2025).

Yang, Qinshan, Jinsong Zhao, and Marvin Rourke. 2019. "Downhole Leak Detection: Introducing a New Wireline Array Noise Tool." *Society of Petroleum Engineers - SPE/ICoTA Well Intervention Conference and Exhibition 2019*.

Zhang, Qiang, Jihong Liu, and Xu Chen. 2024. "A Literature Review of the Digital Thread: Definition, Key Technologies, and Applications." *Systems* **12**(3).